

Project Development Phase
Model Performance Test

Date	11 September 2026
Team ID	NA
Project Name	Metro Ticket Generating System in ServiceNow
Maximum Marks	

Model Performance Testing:

S.No.	Parameter	Screenshot / Values
1.	Data Rendered	• Data Source: ServiceNow Enterprise Instance (Personal Developer Instance) • Primary Tables: sc_request, sc_req_item, u_metro_station_details • Metrics Rendered: Total daily tickets generated, active route distributions, single vs. return journey ratios, fare collection aggregates, and QR scan audit logs.
2.	Data Preprocessing	• Null Value Treatment: Handled unassigned passenger counts by enforcing a default floor of 1. • String Standardization: Trimmed whitespace and standardized station naming conventions from u_metro_station_details. • Type Casting: Converted ticket amounts and journey flags (type_of_journey) from string objects to numeric integers/decimals for accurate aggregation. • Timestamp Normalization: Formatted creation dates into standard UTC and local transit operating hours.
3.	Utilization of Data Filters	• Station Slicers: Interactive filters allowing operators to isolate specific origin and destination nodes (e.g., Ameerpet, Madhapur, LB Nagar, Kukatpally). • Journey Type Slicer: Toggle between Single Journey (1) and Return Journey (2). • Date Range Slicer: Dynamic time filtering for peak-hour analysis (Morning 08:00–11:00 vs. Evening 17:00–20:00). • Payment Mode Filter: Segment data by UPI, Credit/Debit Card, Cash, or Others.
4.	DAX Queries Used	• Total Revenue Generated: <code>Total_Revenue = SUMX(sc_req_item, sc_req_item.price)</code> • Total Tickets Sold: <code>Total_Tickets = DISTINCTCOUNT(sc_req_item.sys_id)</code> • Average Fare per Trip: <code>Avg_Fare = AVERAGE(sc_req_item.price)</code> • Peak Hour Volume Index: <code>Peak_Volume = CALCULATE(COUNT(sc_request.sys_id), HOUR(sc_request.sys_created_on) >= 8 && HOUR(sc_request.sys_created_on) <= 11)</code>
5.	Dashboard design	Number of Visualizations / Graphs: 6 1. <i>KPI Card</i>: Total Daily Tickets Generated 2. <i>KPI Card</i>: Total Revenue Collection (₹) 3. <i>Bar Chart</i>: Top 5 Most Frequent Origin/Destination Routes

		4. <i>Line Chart</i> : Hourly Passenger Throughput (Peak vs. Off-Peak) 5. <i>Donut Chart</i> : Payment Mode Share (UPI vs. Cards vs. Others) 6. <i>Gauge Card</i> : Gate Turnstile QR Scan Success Rate (99.8%)
6.	Report Design	Number of Visualizations / Graphs: 4 1. <i>Executive Summary Grid</i> : Monthly Transit Trends & Revenue Growth 2. <i>Detailed Tabular Matrix</i> : Station-wise Passenger Volume & Fare Breakdown 3. <i>Area Chart</i> : Daily Commuter Surge Comparison (Single vs. Return Trips) 4. <i>Audit Log Table</i> : Failed Scan & System Exception Logs for Station Controllers